AGU PUBLICATIONS

JGR

Journal of Geophysical Research: Space Physics

RESEARCH ARTICLE

10.1002/2017JA024619

Key Points:
- Methods to determine *L-M-N* current sheet coordinates are described and tested
- Quality of results is judged by eigenvalue ratios and consistency using different data intervals and methods and with the geophysical context
- For the interval examined here, the uncertainty of the normal direction was at least several degrees but probably less than 10°

Supporting Information:
- Supporting Information S1

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Citation:
Denton, R. E., Sonnerup, B. U. Ö., Russell, C. T., Hasegawa, H., Phan, T.-D., Strangeway, R. J., et al. (2018). Determining *L-M-N* current sheet coordinates at the magnetopause from Magnetospheric Multiscale data. *Journal of Geophysical Research: Space Physics*, *123*, 2274–2295. https://doi.org/10.1002/2017JA024619

Received 24 JUL 2017
Accepted 14 FEB 2018
Accepted article online 20 FEB 2018
Published online 23 MAR 2018

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Determining *L-M-N* Current Sheet Coordinates at the Magnetopause From Magnetospheric Multiscale Data

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Abstract We discuss methods to determine *L-M-N* coordinate systems for current sheet crossings observed by the Magnetospheric Multiscale (MMS) spacecraft mission during ongoing reconnection, where $\mathbf{e}_L$ is the direction of the reconnecting component of the magnetic field, **B**, and $\mathbf{e}_N$ is normal to the magnetopause. We present and test a new hybrid method, with $\mathbf{e}_L$ estimated as the maximum variance direction of **B** (MVAB) and $\mathbf{e}_N$ as the direction of maximum directional derivative of **B**, and then adjust these directions to be perpendicular. In the best case, only small adjustment is needed. Results from this method, applied to an MMS crossing of the dayside magnetopause at 1305:45 UT on 16 October 2015, are discussed and compared with those from other methods for which $\mathbf{e}_N$ is obtained by other means. Each of the other evaluations can be combined with $\mathbf{e}_L$ from MVAB in a generalized hybrid approach to provide an *L-M-N* system. The quality of the results is judged by eigenvalue ratios, constancy of directions using different data segments and methods, and expected sign and magnitude of the normal component of **B**. For this event, the hybrid method appears to produce $\mathbf{e}_N$ accurate to within less than 10°. We discuss variance analysis using the electric current density, **J**, or the $\mathbf{J} \times \mathbf{B}$ force, which yield promising results, and minimum Faraday residue analysis and MVAB alone, which can be useful for other events. We also briefly discuss results from our hybrid method and MVAB alone for a few other MMS reconnection events.

Plain Language Summary We discuss methods for determining coordinate systems in order to study magnetic reconnection events at the magnetopause, the boundary between the ionized gas in the region of space dominated by the Earth's magnetic field and the ionized gas coming from the solar wind. We introduce a new method that combines results from multiple methods in order to determine the three coordinate directions in space. We demonstrate this method by applying it to an event observed by the Magnetospheric Multiscale spacecraft on 16 October 2015 and at other times.

1. Introduction

Owing in large part to new data from the National Aeronautics and Space Administration's (NASA) Magnetospheric Multiscale (MMS) mission (Burch et al., 2015), a great deal of progress is being made in understanding the processes involved in magnetic reconnection (Burch et al., 2016). MMS has detected thousands of magnetopause crossings (Burch & Phan, 2016), and many papers are being written describing individual reconnection events (see, e.g., the 2016 Geophysical Research Letters special issue, "First results from NASA's MMS Mission," and the 2017 Journal of Geophysical Research special collection, "Magnetosphere Multiscale (MMS) mission results throughout the first primary mission phase").

Table 1
Analysis Methods

Acronym	Description	Relevant eigenvector[a]	For direction	Reference
MVAB	Variance of **B**	max	*L*	Denton et al. (2016a)
		min	*N*	Sonnerup and Scheible (1998)
MVABC	Constrained variance of **B**	int[b]	*N*	Sonnerup and Scheible (1998)
MVAJ	Variance of **J**[c]	min	*N*	Haaland et al. (2004)
MVAJxB	Variance of $\mathbf{J} \times \mathbf{B}$	max	*N*	this paper
MDDB	Directional derivative of **B**	max	*N*	Shi et al. (2005)
MFR	Minimum Faraday residue	min	*N*	Terasawa et al. (1996), with Khrabrov and Sonnerup (1998)

[a]"max," "int," and "min" indicate the maximum, intermediate, and minimum eigenvalues, respectively. [b]Here "int" refers to the direction of minimum variance in the plane perpendicular to $\langle \mathbf{B} \rangle_t$, where $\langle \rangle_t$ indicates a time average over the data set used. [c]**J** is the electric current density calculated from particle data.

The simplest description of the magnetic reconnection process is two-dimensional, that is, its structure is confined to a "reconnection plane." At the magnetopause, this plane contains the *L* axis along the direction of the component of the magnetic field, **B**, that reverses sign (the "reconnecting field," positive northward) and, perpendicular to it, the *N* axis perpendicular to the current sheet (the "normal component," positive away from Earth). The *M* axis, positive westward, completes the right-handed orthogonal triad. In the usual two-dimensional model the invariant direction and the reconnection line (the X line) are along *M*. An *M* component of the magnetic field (a guide field) may also be present, along with the all-important reconnection electric field component. However, for the asymmetric conditions typical at the magnetopause, the invariant direction and the X line may be differently oriented (for a brief overview, see Liu et al., 2015). This *L-M-N* system is similar to, but not identical to, the boundary normal system defined by Russell and Elphic (1979). Our presentation will be directed to magnetopause events but is in large part also applicable to events in the geomagnetic tail and elsewhere.

Many mathematical models and computer simulations of reconnection, as well as various methods for the reconstruction of their structure in the reconnection plane from measured data (e.g., Sonnerup et al., 2016), are based on the two-dimensional assumption. For a recent MMS application, see Hasegawa et al. (2017). An accurately determined *L-M-N* system is indispensable for the interpretation of such events. In particular, a nonzero *N* component of the magnetic field indicates the presence of reconnection: Its sign indicates the location relative to the reconnection site, and its magnitude relative to the reconnecting magnetic field component, along with the magnitude of the *M* component of the electric field, gives indications of the reconnection rate. The change of the reconnecting field across the magnetopause, along with the behavior of the guide field, is also an important parameter in helping to establish the overall characteristics and geometry of the reconnection process. To determine spatial scales and perform reconstruction, the motion of the structure is also required. Determining this motion from measured data is a complicated issue that we will discuss elsewhere. (For a recent MMS-based approach, see Denton et al., 2016a.)

It is the main purpose of the present paper to examine in detail a "hybrid method" to obtain a reliable determination of the *L-M-N* orientation, as recently proposed by Denton et al. (2016a). It is a combination of the minimum directional derivative (MDD) method developed by Shi et al. (2005), which requires data from the four spacecraft tetrahedron observation platform provided by MMS and the much older, single-spacecraft method called minimum variance analysis of **B** (MVAB) (Sonnerup & Cahill, 1967; Sonnerup & Scheible, 1998). Acronyms of methods used in this paper are listed in Table 1, along with a description, the relevant eigenvalues and directions obtained, and relevant references, if applicable. MVAB was originally designed to provide a prediction of the *N* direction, but many years of experience have shown that it can fail to give acceptable results for that direction as a consequence of near degeneracy of the magnetic variance matrix in the *M-N* plane. On the other hand, the *L* axis is often well predicted as the direction of maximum variance of the corresponding component of **B**. For MDD used with **B** (MDDB), the situation is different: the maximum gradient direction often provides an accurate prediction of the *N* axis, while there are problems with degeneracy in the *L-M* plane, that is, the plane tangent to the magnetopause. The hybrid method therefore is based on *L* determined from MVAB and *N* determined from MDDB (in both acronyms, the letter "M" may now be taken

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to stand for "maximum" or "minimum," depending on the context). In the best case, the resulting *L* and *N* directions should come out nearly perpendicular. The angular deviation from orthogonality provides one of several relevant measures of quality.

As a test case, we use one event, namely, an outward bound magnetopause crossing of MMS at around 1306 UT on 16 October 2015, which occurred at coordinates (8.32, 8.50, −0.70) R_E in the Geo-Solar Ecliptic (GSE) coordinates used throughout this paper. We examine results from our hybrid method and compare them to those using other methods. This traversal of the magnetopause occurred in the early part of the back-and-forth crossings discussed in the seminal paper by Burch et al. (2016), where it was used to determine their coordinate system. It occurred before the encounter with the electron diffusion region at around 1307 UT (Burch et al., 2016; Denton et al., 2016a, 2016b).

By determining a coordinate system for the current sheet crossing, we do not mean to imply that the relevant physical directions, such as the normal direction across the current sheet, are stationary in time. There are variations, as will be apparent from our results. But our goal is to determine representative directions for the purpose of defining a coordinate system suitable for the study of the reconnection process.

The presentation is organized as follows. In section 2, we summarize the basic data sets to be used and how they were prepared for our analysis. In section 3, we briefly review the methods to be used. Section 4 presents our analysis for the current sheet crossing on 16 October 2015. In section 4.1, we apply our hybrid method using MVAB and MDDB to a 4.5 s time interval, centered at 1305:45 UT, to generate a representative *L-M-N* coordinate system. These results are compared to those from MVAB using a much longer time period (Burch et al., 2016). In addition, we consider results using other time intervals nested around 1305:45 UT. In section 4.2, we present other determinations of the *N* direction from: MVAB; MVAJ using the electric current density, **J**; MVAJxB using $\mathbf{J}\times\mathbf{B}$; and the minimum Faraday residue (MFR) technique using the electric field (Terasawa et al., 1996). Section 5 contains a discussion of how the normal vector orientation from our hybrid method compares to those found from magnetopause models. A summary and discussion of the significance of our results is given in section 6.

2. Data

The four MMS spacecraft had a nominal separation of about 15 km, with the relative positions thought to be known to an accuracy of about 20 m (T. Williams, private communication, 2017).

We used MMS FluxGate Magnetometer data (Russell et al., 2014), Electric Double Probe (EDP) data (Ergun et al., 2016; Lindqvist et al., 2016), and ion and electron bulk velocity moments from the Fast Plasma Instrument (FPI) (Pollock et al., 2016). The data were downloaded at burst mode resolution, 7.8 ms for the magnetic field from FluxGate Magnetometer, 7.3 ms for the electric field from Electric Double Probe, and 150 ms for the ions and 30 ms for the electrons from FPI. All data were converted to the resolution of the electron data (30 ms) using common times for all four spacecraft that were chosen so as to minimize the shift in the individual times of the electron data. The electron and ion velocities were interpolated, and the magnetic field and electric field data were averaged, to the new times using linear weighting. The result was a data set with all the data sampled at the same times with 30 ms resolution.

Some methods that we will discuss below use the electric current density calculated from the particle moments, $\mathbf{J} = e\left(n_{\text{ion}}\mathbf{V}_{\text{ion}} - n_{\text{electron}}\mathbf{V}_{\text{electron}}\right)$, where *e* is the proton charge and n_s and $\mathbf{V}_s$ are the density and velocity moments for species *s* from FPI. (For simplicity, we have simply used the MMS ion and electron fluxes, ignoring the minute presence of He+2 and other species.) We also calculated the current density $\mathbf{J}_{\text{curl}}$, from the magnetic field **B** measured by all four spacecraft using the "curlometer technique" (Robert et al., 1998), but use of the particle current density led to better results (a larger separation of eigenvalues and more consistent directions; see section 4.2.3).

Only the MFR method (Khrabrov & Sonnerup, 1998; Terasawa et al., 1996) requires the electric field, **E**. We tried using the measured electric field but found more consistent results from different analysis intervals using $\mathbf{E} = -\mathbf{V}_{\text{electron}} \times \mathbf{B}$.

3. Methods

For all of the methods that we will be considering (see Table 1), a 3 × 3 symmetric matrix **M** is calculated for a particular time interval. Then the eigenvector problem for **M** is solved, yielding three positive definite

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eigenvalues along with three corresponding orthogonal directions. The three eigenvalues are labeled λ_1, λ_2, and λ_3, in decreasing order (λ_1 largest). Some of the methods were described by Sonnerup et al. (2006).

There are four considerations that we will use to evaluate whether the results of these methods are reliable. First of all, a large ratio of λ_1/λ_2 suggests that the estimation of the maximum eigenvalue direction should be good, and a large ratio of λ_2/λ_3 suggests that the estimation of the minimum eigenvalue direction should be good. If two eigenvalues are close to the same value, the matrix is close to being degenerate, and the corresponding eigenvectors may not be well determined.

Second, we look for consistency in the direction as we vary the time interval used to evaluate the direction.

Third, ideally, we will be able to verify the results using more than one method.

Fourth, we look for the expected sign and a reasonable (i.e., not too large) magnitude of the normal magnetic field component in the reconnection event being studied.

For minimum variance of the magnetic field, Sonnerup and Scheible (1998) state that, as a rough rule of thumb, the eigenvalues should be separated by at least a factor of 10 for the directions to be well determined. However, we emphasize that a ratio above 10 does not always guarantee that the direction obtained is near the desired true direction. Similarly, a ratio below 10 does not necessarily mean that the direction obtained must be far from the true answer. What the eigenvalue ratio does is to provide a simple qualitative measure of the uncertainty of the direction obtained. In this sense, a ratio of 10 or larger is an a priori indication of good quality. In practice, the situation is often complex, which means that, regardless of eigenvalue ratio, the direction obtained must be examined in its geophysical context, including a host of other observations and model expectations.

The accuracy needed will depend on the particular application, but generally, we consider an accuracy of 5° to be good, an accuracy of 10° to be adequate for some purposes, and an accuracy of 20° or more to be poor. Note that, with a total field of say 30 nT, an error of 5.7° in the normal vector would lead to an error of 3 nT in the normal field component, which is comparable to actual values during reconnection events. In other words, to obtain a reliable value of this field component, an extremely accurate determination of the normal direction is required, which may or may not be achievable in some events.

3.1. MVA, Variance of a Vector Field

For minimum (or maximum) variance of the magnetic field **B** (MVAB) (Sonnerup & Scheible, 1998, and references therein), the variance matrix $\mathbf{M}^{\mathrm{MVAB}} = \mathbf{M}^{BB}$ is calculated, where $M^{BB}_{ji} = \langle \delta B_j \delta B_i \rangle_t$, $\delta B_i = B_i - \langle B_i \rangle_t$ is the i'th component of the perturbed magnetic field and $\langle \rangle_t$ denotes a time average over an interval of measurements. Although MVAB was originally developed for data from a single spacecraft, it can be used for multiple spacecraft data, either by doing the calculation separately for each spacecraft, by concatenating the data from all spacecraft (as we have done here), or by first averaging the data from the spacecraft at each time. In our events, the results from these different approaches are very similar. The maximum variance eigenvector of $\mathbf{M}^{\mathrm{MVAB}}$ is expected to be in the direction of the reversing reconnection magnetic field, the L direction. For many magnetic reconnection events, this direction is the most well determined.

In the one-dimensional approximation with variation only across the current sheet, $\nabla \cdot \mathbf{B} = 0$ leads to zero variation of the magnetic field component for that direction. So, if the structure is nearly one-dimensional and the two smallest eigenvalues from MVAB are well separated, then the minimum variance direction would usually correspond closely to $\mathbf{e}_N$. For an exception see (Sonnerup et al., 2008, page 11).

In some cases, current sheet crossings are close to tangential discontinuities, with only a small normal component of **B**. In such a case, one can use the constrained minimum variance of **B** method (MVABC) (Sonnerup & Scheible, 1998) in order to get an approximate normal direction. In this method, the direction along the average magnetic field is projected out of the matrix. A modified matrix $\mathbf{M}^{\mathrm{MVABC}} = \mathbf{P} \cdot \mathbf{M}^{BB} \cdot \mathbf{P}$ is constructed, where $P_{ji} \equiv \delta_{ji} - b_j b_i$, δ_{ji} is the Kronecker delta and $\mathbf{b} \equiv \langle \mathbf{B} \rangle_t / \langle |\mathbf{B}| \rangle_t$. The minimum eigenvalue of $\mathbf{M}^{\mathrm{MVABC}}$ will be zero, with the corresponding eigenvector in the direction of the average magnetic field. The eigenvector of the intermediate eigenvalue is the constrained minimum variance direction. The average of the magnetic field component in this direction is precisely zero; so one disadvantage of this method is that it provides no information about the sign and magnitude of the magnetic field component in this direction (as well as in the maximum variance direction).

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We will also show results for analysis using the variance of **J** (MVAJ) (Haaland et al., 2004). Since $\nabla \cdot \mathbf{J} \approx 0$, the minimum variance direction of **J** for a nearly one-dimensional structure, for which $\lambda_{\mathrm{MVAJ},2}/\lambda_{\mathrm{MVAJ},3}$ is large, should again correspond to $\mathbf{e}_N$.

Additionally, we will show results for analysis using the variance of the magnetohydrodynamic force $\mathbf{J} \times \mathbf{B}$ (MVAJxB). Since **J** is essentially perpendicular to the normal direction and since B_N is small in typical reconnection events, by far the largest component of $\mathbf{J} \times \mathbf{B}$ is in the direction normal to the reconnecting current layer. Because the force is zero on the two sides far from the current sheet, its largest variance also is in the normal direction.

3.2. MDD, Directional Derivative of a Vector Field

Minimum (or maximum) directional derivative of **B** (MDDB) analysis (Denton et al., 2010, 2012; Shi et al., 2005) requires four spacecraft to calculate the directional derivative, or gradient, of **B**. One starts with the matrix $\partial_j B_i$ that is the j'th component of the gradient of B_i. This matrix is multiplied by its transpose in order to produce a symmetric matrix, $\mathbf{M}^{\mathrm{MDDB}}$, which is then diagonalized to find its eigenvalues and eigenvectors. For many magnetic reconnection events, the maximum gradient direction, corresponding to $\mathbf{e}_N$, is the best determined eigenvector because of the large gradient across the current sheet.

We might expect that the minimum gradient direction would be in the out of plane direction, $\mathbf{e}_M$. Denton et al. (2016a) showed, however, that for the event observed by MMS at 1307 UT on 16 October 2015, the minimum gradient was actually in the L direction.

In principle, MDDB can give the directions at each point in time, since the gradient of **B** can be calculated using the instantaneous magnetic field. We will use the instantaneous direction to get an estimate of the statistical error (due to different measurements and time variation) for the one event that we will analyze in detail. But for the purposes of calculating a single direction for an event, we will sum the point by point matrices from MDDB into one total matrix before finding the eigenvalues and eigenvectors (Denton et al., 2010; Denton et al., 2012). (Another approach is to use the median or average of the time-dependent eigenvectors Denton et al., 2010.)

We will discuss, but not present in detail, results of MDD using other vector fields.

3.3. MFR Method

The MFR method minimizes the differences between the fields predicted by Faraday's law and those observed for a moving one-dimensional time-independent structure (Terasawa et al., 1996). Khrabrov and Sonnerup (1998) formulated the solution as a matrix problem yielding a unique answer. A matrix $\mathbf{M}^{\mathrm{MFR}} = \langle|\delta\mathbf{E}|^2\rangle_t \delta_{ji} - M^{EE} - P_j P_i / \langle|\delta\mathbf{B}|^2\rangle_t$ is constructed, where $\mathbf{P} = \langle\delta\mathbf{E}\times\delta\mathbf{B}\rangle_t$. This method predicts that $\mathbf{e}_N$ should be the minimum eigenvalue direction.

4. The 16 October 2015 Event

Burch et al. (2016) analyzed a reconnection event observed by MMS on 16 October 2015 at 1307:02 UT. To determine the coordinate system, however, they used an earlier interval that had a more complete crossing of the magnetopause, 1305:40–1306:09, shown in Figure 1. (Since there continued to be a significant gradient of **B** within the magnetosheath, this magnetopause crossing may also not be complete.) Using MVAB with the full interval shown in the figure, Burch et al. (2016) found that $\mathbf{e}_L = (0.3665, -0.1201, 0.9226)$, $\mathbf{e}_M = (0.5694, -0.7553, -0.3245)$, and $\mathbf{e}_N = (0.7358, 0.6443, -0.2084)$. Figure 1a shows the L, M, and N components of $\mathbf{B}_{\mathrm{av}}$ using this coordinate system for the entire time interval with time measured from 1305 UT. Here and elsewhere, the subscript "av" indicates an average over the four spacecraft.

The eigenvalues from MVAB were 500 nT^2, 32.0 nT^2, and 13.0 nT^2, so $\lambda_{\mathrm{MVAB},1}/\lambda_{\mathrm{MVAB},2} = 15.6$ and $\lambda_{\mathrm{MVAB},2}/\lambda_{\mathrm{MVAB},3} = 2.5$. While the maximum variance direction, corresponding to L, appears to be adequately defined by this procedure, the minimum variance direction, corresponding to N, is open to question. (However, our analysis has shown that the normal direction used by Burch et al., 2016 was adequate to describe the 1307:02 UT crossing.) The fact that the variations in $B_{\mathrm{av},N}$ are of comparable amplitude to the variations in $B_{\mathrm{av},M}$ is clear from Figure 1a. Because of this, we will use the hybrid method rather than the MVAB eigenvector triad to determine the coordinate system.

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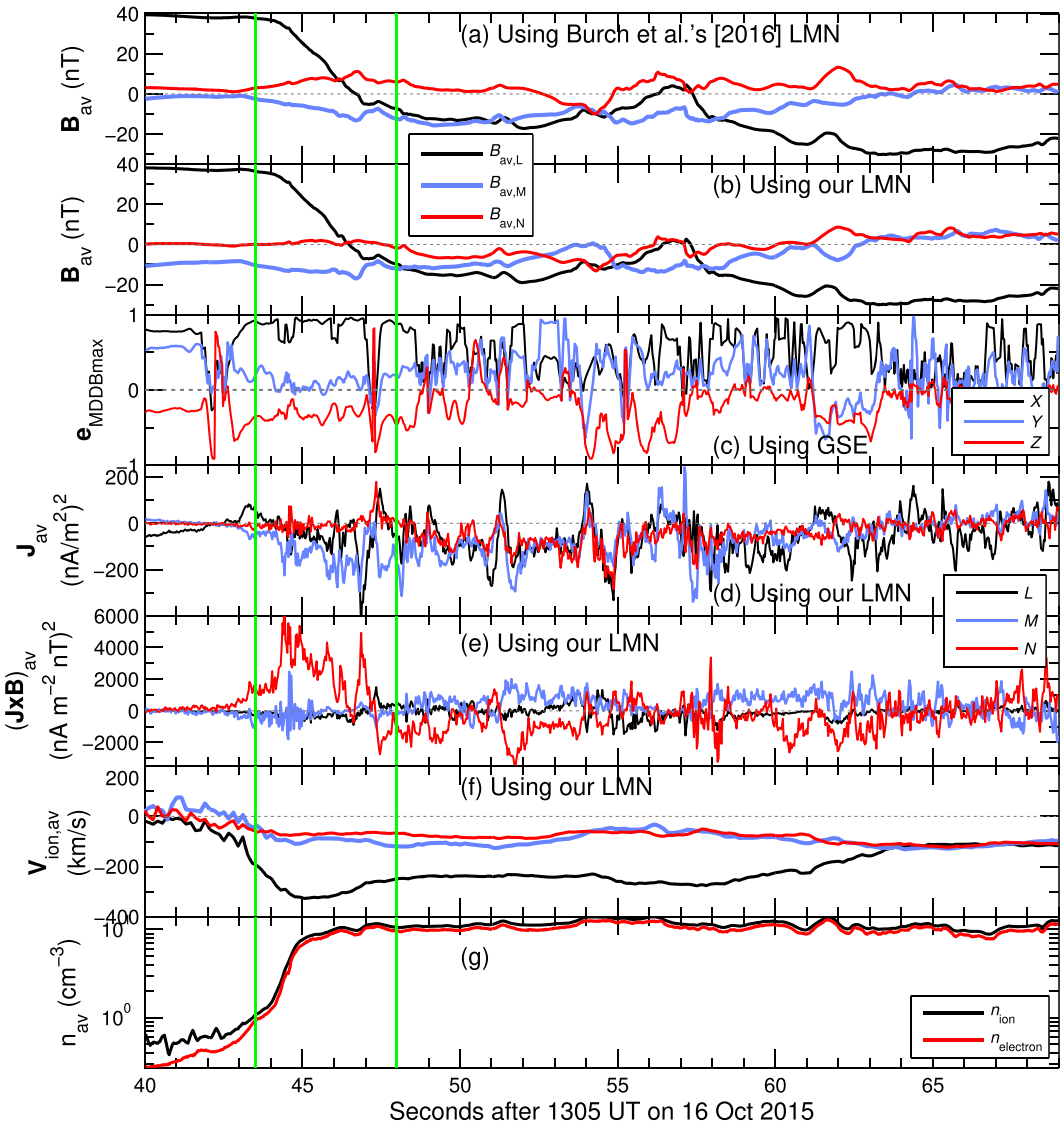

Figure 1. Time-dependent data for the event at 1305 UT on 16 October 2015. (a) *L*, *M*, and *N* components of **B** averaged over the four spacecraft, $\mathbf{B}_{av}$, using the Burch et al. (2016) coordinate system based on variance of **B** using the entire time interval. (b) Same as Figure 1a but using our coordinate system for the 4.5 s time interval between the green vertical lines. (c) Direction cosines to the Geo-Solar Ecliptic (GSE) axes for the maximum gradient direction from maximum directional derivative of **B**. (d–f) The *L*, *M*, and *N* components using our hybrid coordinate system of the following quantities averaged over the four spacecraft: (d) the current density, $\mathbf{J}_{av}$; (e) the average magnetohydrodynamic force, $(\mathbf{J}\times\mathbf{B})_{av}$; (f) the ion velocity, $\mathbf{V}_{ion,av}$; and (g) the ion and electron densities, $n_{ion,av}$ and $n_{electron,av}$, showing their close agreement, except on the left in the magnetosphere.

4.1. Hybrid Method Making Use of MDD Analysis

Denton et al. (2016a) found an *L-M-N* coordinate system by combining the maximum variance direction from MVAB with the maximum gradient direction of MDDB. Using this method, Denton et al. (2016a) obtained *L-M-N* directions around 1307:02 UT, close to the time of observation of the electron diffusion region studied by Burch et al. (2016). The normal directions from these two studies differed by 10.9°. Here we will continue to focus on the earlier time interval with a magnetopause crossing at about 1305:45 UT.

The eigenvalues from MDDB for the time interval 1305:40 to 1306:09 were $\lambda_{MDDB,i} = 0.021$ (nT/km)^2, 0.0059 (nT/km)^2, and 0.0023 (nT/km)^2, yielding $\lambda_{MDDB,1}/\lambda_{MDDB,2} = 3.6$ and $\lambda_{MDDB,2}/\lambda_{MDDB,3} = 2.6$. The resulting maximum gradient eigenvector was $\mathbf{e}_N = (0.8903, 0.3556, -0.2844)$, which deviates from the MVAB

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normal for the same interval (which is the normal used by Burch et al., 2016 to represent the 1307:02 UT normal direction) by 19.4°. However, in this case one cannot be confident about any of the directions from MDDB. A better eigenvalue separation for both MVAB and MDDB can be found by picking a time interval that includes just the region of the largest gradient in the magnetic field. Such an interval may not give results that represent all aspects of the full crossing or of the crossing at 1307:02 UT. But it does include most of the current sheet crossing, including all of the current sheet directly adjoining the magnetosphere, which is less likely to be distorted because the plasma beta is lower in the magnetosphere. And it provides a suitable test case for our hybrid method and its comparison with other methods.

4.1.1. Hybrid Coordinate System Evaluated Using a Shorter Time Interval

Consider the time interval 1305:43.5 to 1305:48 UT marked off by the green vertical lines in Figure 1. For this interval, the eigenvalues for magnetic variance analysis are better separated, with $\lambda_{\mathrm{MVAB},i}$ equal to 273 nT^2, 3.75 nT^2, and 0.593 nT^2. The eigenvectors for MVAB and MDDB are listed for this interval in Text S1 of the supporting information. Based on the eigenvalue ratio, the determination of the minimum variance direction is better for this shorter interval, with $\lambda_{\mathrm{MVAB},2}/\lambda_{\mathrm{MVAB},3} = 6.3$, compared to 2.5 for the longer interval. But it still does not satisfy our goal for a value exceeding 10. The maximum variance direction is also much better determined, with $\lambda_{\mathrm{MVAB},1}/\lambda_{\mathrm{MVAB},2}$ having increased from 15.6 to 73. The eigenvalues from MDDB, $\lambda_{\mathrm{MDDB},i}$, are 0.042 (nT/km)^2, 0.0035 (nT/km)^2, and 0.0018 (nT/km)^2, yielding $\lambda_{\mathrm{MDDB},1}/\lambda_{\mathrm{MDDB},2} = 12$ and $\lambda_{\mathrm{MDDB},2}/\lambda_{\mathrm{MDDB},3} = 1.9$. While the determination of the minimum gradient direction remains poor, the determination of the maximum gradient direction now appears adequate.

Figure 1c shows the maximum gradient direction from MDDB, determined at each time. For most of the time interval between the green vertical lines, this direction is roughly consistent, albeit with some brief significant variations, particularly just after 47 s. (At that time, the motion of the spacecraft in the normal direction is stagnant and there is some significant time variation in the relative spacecraft magnetic field values; see Figure S1.) But it must be kept in mind that, with a cadence of 30 ms, MDDB provides nearly instantaneous estimates of the normal direction, which on short time scales can undergo large variations caused by local fluctuations. By contrast, most standard single-spacecraft methods give a single estimate for the time interval used, in this case 4.5 s. In order to get an estimate of the *N* direction for the entire time interval, we will use the normal direction from the MDDB matrix averaged over this interval. The resulting MDDB maximum gradient eigenvector forms an angle of 87.1° with the MVAB maximum variance eigenvector from the same interval, so the two directions are nearly orthogonal. We will use the MVAB maximum variance eigenvector and the MDDB maximum gradient eigenvector to construct our *L*-*M*-*N* system; for brevity, they will be referred to as MVABmax and MDDBmax.

We now denote MVABmax by $\mathbf{e}_{L'}$ and MDDBmax by $\mathbf{e}_{N'}$ and then construct our positive *M* direction as being along $\mathbf{e}_{N'} \times \mathbf{E}_{L'}$. Rather than making $\mathbf{e}_L$ exactly equal to $\mathbf{e}_{L'}$ or $\mathbf{e}_N$ exactly equal to $\mathbf{e}_{N'}$, we implement an intermediate choice by rotating both vectors around $\mathbf{e}_M$. The angles of rotation are chosen so that if the eigenvalue ratio $\lambda_{\mathrm{MVAB},1}/\lambda_{\mathrm{MVAB},2}$ is larger than $\lambda_{\mathrm{MDDB},1}/\lambda_{\mathrm{MDDB},2}$, the rotation of $\mathbf{e}_{L'}$ to $\mathbf{e}_L$ will be less than that of $\mathbf{e}_{N'}$ to $\mathbf{e}_N$. On the other hand, if the eigenvalue ratio $\lambda_{\mathrm{MVAB},1}/\lambda_{\mathrm{MVAB},2}$ is smaller than $\lambda_{\mathrm{MDDB},1}/\lambda_{\mathrm{MDDB},2}$, the rotation of $\mathbf{e}_{L'}$ to $\mathbf{e}_L$ will be greater than that of $\mathbf{e}_{N'}$ to $\mathbf{e}_N$. The procedure is described in Appendix A. In the case of our 4.5 s time interval, $\lambda_{\mathrm{MVAB},1}/\lambda_{\mathrm{MVAB},2} = 73$, which is very large; so $\mathbf{e}_L$ is very close to $\mathbf{e}_{L'}$. The result is $\mathbf{e}_L$ = (0.3894, -0.2938, 0.8730), $\mathbf{e}_M$ =(-0.0079, -0.9488, -0.3157), and $\mathbf{e}_N$ = (0.9210, 0.1160, -0.3718) in GSE. More details are given in Text S1 of the supporting information.

Since our method for obtaining the *L*-*M*-*N* system combines results from MVAB and MDDB, we call it a hybrid method. The resulting *L*, *M*, and *N* directions for this current sheet crossing at 1305:45 deviate by 10°, 35°, and 34°, respectively, from those used by Burch et al. (2016), a large difference. Figure 1a shows *L*, *M*, and *N* components of $\mathbf{B}_{\mathrm{av}}$ for the 16 October 2015 event using the Burch et al. (2016) coordinate system from MVAB for the entire interval, while Figure 1b gives those components along our hybrid coordinates derived using the 4.5 s interval. The *L* components in the two panels are similar, but the *M* and *N* components are significantly different. In the 4.5 s interval between the green lines, the B_N component in Figure 1b is approximately constant as expected in a one-dimensional structure. This behavior occurs even though MDDB does not involve explicit minimization of the variance of B_N. On the other hand, in Figure 1a this variance was minimized (via MVAB) but now over the full interval. The resulting variance of B_N in Figure 1a between the green lines and elsewhere is substantial, indicating that some aspect of the assumption of a time-independent, one-dimensional structure of fixed orientation must be violated.

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Table 2
Direction Notation

Label	Definition
L, M, N	Either the specific directions from our hybrid method with MDDB using the 4.5 s interval or generic directions
L'	Interval-dependent MVAB maximum eigenvalue direction
N'	Interval-dependent N direction using any of the methods in Table 1
L^*, M^*, N^*	Interval-dependent L, M, and N after adjusting L' and N' to be perpendicular using the method of Appendix A

Note. MDDB = directional derivative of **B**; MVAB = variance of **B**.

4.1.2. Direction Notation

From now on, the labels L, M, and N will usually indicate the directions calculated using our hybrid scheme for the 4.5 s interval. (Occasionally they may also be used in their generic sense.) Below we will consider coordinate directions calculated using varying time intervals with differing methods to find the N direction. The interval-dependent and method-dependent coordinate labels will be L^*, M^*, and N^*. As was the case for our hybrid method using MVABmax and MDDBmax, some adjustment of the raw directions may be necessary to make $\mathbf{e}_{L^*}$ and $\mathbf{e}_{N^*}$ orthogonal to each other. The label associated with the unadjusted MVABmax eigenvector (used throughout the paper as the starting point for calculating the L direction) will be L', while the label associated with the unadjusted direction used to get the N direction will be N'. If the MVAB minimum variance (MVABmin) direction is used for $\mathbf{e}_{N'}$, then $\mathbf{e}_{N^*} = \mathbf{E}_{N'}$ because $\mathbf{e}_{N'}$ is already orthogonal to $\mathbf{e}_{L'}$. But for other methods, adjustment of both $\mathbf{e}_{L'}$ and $\mathbf{e}_{N'}$, as described in Appendix A, is required. Then $\mathbf{e}_{L^*}$ will not be exactly equal to $\mathbf{e}_{L'}$, and $\mathbf{e}_{N^*}$ will not be exactly equal to $\mathbf{e}_{N'}$. We summarize our notation for directions in Table 2.

4.1.3. Estimates of Accuracy

The statistical error of the MVABmax eigenvector for the 4.5 s interval, using equation 8.23 of Sonnerup and Scheible (1998), is very small, about 0.2° and does not provide a useful measure of actual accuracy. To get such a measure, we examine how the direction changes using different time intervals.

Figure 2 shows results of MVAB using a nested set of time intervals of length Δt_{range} centered on 1305:45.75 UT. The default time interval Δt_0 is 4.5 s, the small interval that we used in section 4.1.1. Figure 2a shows the maximum, intermediate, and minimum eigenvalues of $\mathbf{M}^{\text{MVAB}}$. The black, blue, and green curves in Figure 2b are the direction cosines to the X, Y, and Z GSE coordinate axes, respectively, for the MVABmax eigenvector, $\mathbf{e}_{\text{MVABmax}}$. We expect that the MVABmax eigenvector should correspond closely to the L direction. Therefore, in Figure 2b we also show as horizontal dotted lines the direction cosines for the L direction from the hybrid method using the 4.5 s interval (hence, the notation "$\mathbf{e}_L$ dotted" next to the panel label "(b)"). The intermediate and minimum eigenvalue eigenvectors might correspond to the M and N directions, respectively. So in Figure 2c, we show the direction cosines to the intermediate eigenvalue eigenvector (curves) along with direction cosines to our M direction (dotted horizontal lines), and in Figure 2d we show the direction cosines to the MVABmin eigenvector (curves) along with direction cosines to our N direction (dotted horizontal lines).

Only the MVABmax eigenvector is relatively constant across different time intervals. (Compare the constancy of the curves in Figure 2b to the irregularity of the curves in Figures 2c and 2d.) In other words, there is a large rotation of the intermediate and minimum variance axes around the maximum variance axis as Δt_{range} is varied. The detailed physical reason for this rotation is not clear, but it is related to the fact that the separation of the two smallest eigenvalues is poor (Figure 2a). There is a much larger separation between the maximum and intermediate eigenvalues, indicating that the maximum variance direction is the one best determined. In Figure 2 and subsequent similar figures, panels with gray background indicate eigenvector directions that are poorly determined or less useful because they cannot be easily interpreted.

The different time intervals are sampling different kinds of structure. We will have the greatest confidence in the L direction if we get the same result for any time interval. But at the smallest time intervals, we may be investigating waves or other "substructure" smaller than that of the overall reconnection structure. So the rough constancy of the L direction as we transition to the smallest intervals is not necessarily required to find a good L direction. Similarly, on the longest time scales, there may be time-dependent changes in the structure orientation due to the global dynamics. Therefore, not all time intervals are equally valuable. Here we chose

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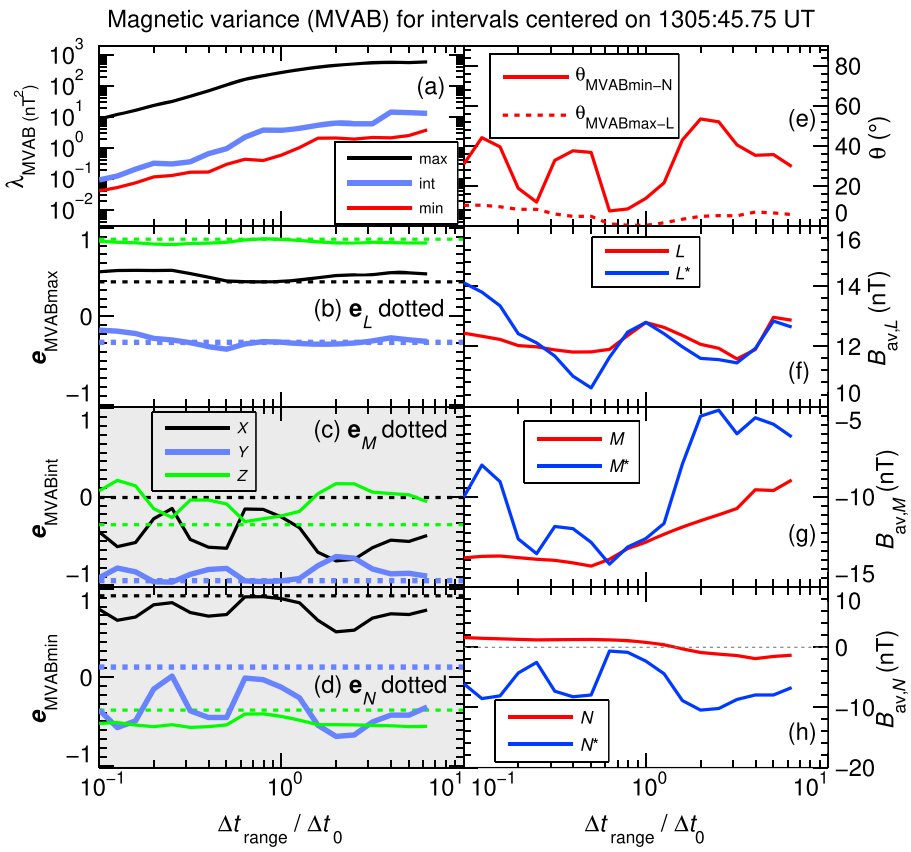

Figure 2. Results of variance analysis of **B** (maximum variance direction of **B** (MVAB)) for different nested time intervals normalized to the $\Delta t_0 = 4.5$ s interval between the vertical green lines in Figure 1. (a) Eigenvalues. (b–d) Direction cosine curves relative to Geo-Solar Ecliptic for the eigenvectors of maximum, intermediate, and minimum variance, with color code shown in panel (c). The horizontal dotted lines in Figures 2b, 2c, and 2d are drawn at the direction cosines for our *L*, *M*, and *N* directions, respectively. Gray panels signify poorly determined eigenvector directions. (e) Angle between the MVABmin eigenvector and our normal vector $\mathbf{e}_N$ (red solid curve), and between the MVABmax eigenvector and our $\mathbf{e}_L$ (dotted red curve). (f–h) Time-averaged *L*, *M*, and *N* components of **B** for each time interval, relative to our fixed LMN system (red curves) and relative to the interval-dependent L^*-M^*-N^* system (blue curves) composed of the MVAB eigenvectors.

the default interval based on Figure 1 because that interval includes a significant portion of the magnetopause crossing (1305:43.5 to 1305:48 UT) but omits the extended period of back-and-forth (time-dependent) motion on the magnetosheath side of the current sheet (1305:48 to 1305:62 UT).

We always use the MVABmax eigenvector (curves in Figure 2b) for $\mathbf{e}_{L'}$. In Figure 2e, the red dotted curve is the angle between the interval-dependent $\mathbf{e}_{L'}$ and our hybrid $\mathbf{e}_L$ determined using the 4.5 s interval ($\Delta t_{\text{range}}/\Delta t_0 = 1$). This angle is within about 10° for all the time intervals. It is not exactly zero even at $\Delta t_{\text{range}}/\Delta t_0 = 1$ because there was a slight adjustment of MVABmax from the 4.5 s interval to get $\mathbf{e}_L$, as described in section 4.1.1 and Appendix A. For nested time intervals up to a factor of 2.5 times smaller or 3.2 times larger than the 4.5 s interval, the maximum variance direction is within 5.2° of our $\mathbf{e}_L$ (for the 4.5 s interval). Using intervals up to a factor of 6.3 larger, the maximum variance direction is within 7.0° of our $\mathbf{e}_L$. Considering that the larger intervals use data that may include temporal changes and that small time intervals might not yield as accurate results because of substructure or bad statistics, an uncertainty of about 5°, based on this variation, seems reasonable.

The other curves in Figures 2e and 2f–2h relate to the use of the MVABmin eigenvector to determine the *N* direction and will be discussed later in section 4.2.1.

The standard deviation of the angle between our $\mathbf{e}_N$ from the 4.5 s interval and the instantaneous maximum gradient directions in that interval (Figure 1c), after making them orthogonal to $\mathbf{e}_L$, was 22°. But after dividing by the square root of the number of measurements, the statistical uncertainty of the mean value

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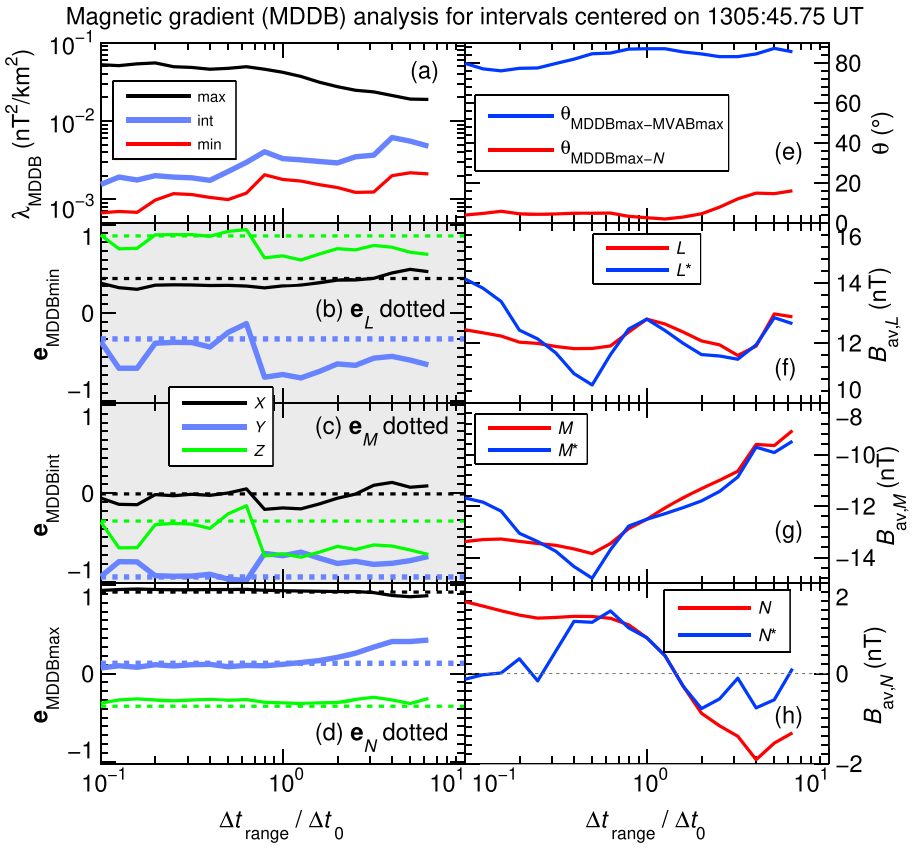

Figure 3. Results from directional derivative analysis of **B** (MDDB)). Similar to Figure 2 but now showing (a) eigenvalues from MDDB. Direction cosines for the (b) minimum, (c) intermediate, and (d) maximum gradient directions (curves) plotted along with the direction cosines of our *L*, *M*, and *N* directions, respectively (horizontal dotted lines). (e) Angles between the MDDBmax eigenvector and the MVABmax eigenvector (blue curve) or our *N* direction (red curve). Time-averaged interval-dependent (f) *L*, (g) *M*, and (h) *N* components calculated using our fixed coordinate system (red curves) or an interval-dependent L^*-M^*-N^* coordinate system combining the MVABmax eigenvector for $\mathbf{e}_{L'}$ with the MDDBmax eigenvector for $\mathbf{e}_{N'}$ (blue curves).

was only 1.8°. Note also that the maximum gradient direction found from the median of the time-dependent eigenvectors was only 1.4° from the maximum gradient direction determined using the average gradient matrix, demonstrating consistency of results from these two possible versions of the method.

To examine the stability of the MDDB results using different time intervals, we plot in Figure 3 the eigenvalues and eigenvectors for MDDB using a format similar to that used for MVAB in Figure 2. As was the case for the MVABmax eigenvector in Figure 2b, the MDDBmax eigenvector in Figure 3d is nearly constant over a large range of time intervals. The blue curve in Figure 3e shows $\theta_{\text{MDDBmax-MVABmax}}$, the angle between the MDDBmax eigenvector and the MVABmax eigenvector. This angle is close to 90°, indicating near orthogonality, except for the smallest time intervals for which $\theta_{\text{MDDBmax-MVABmax}}$ drops to as low as 76°. The red curve in Figure 2e shows $\theta_{\text{MDDBmax-}N}$, the angle between the MDDBmax eigenvector and our *N* direction. For $\Delta t_{\text{range}}/\Delta t_0$ up to 2, $\theta_{\text{MDDBmax-}N}$ is within 5.8° of zero. Thus, again, an uncertainty of about 5° for a representative normal direction seems reasonable as long as the interval is not too big.

4.1.4. Average Magnetic Field for Different Intervals

The blue curves in Figures 3f–3h show, respectively, the L^*, M^*, and N^* components of the magnetic field averaged over the four spacecraft and averaged over each time interval. The L^*-M^*-N^* coordinate system is constructed for each interval using the hybrid method of Appendix A with the MVABmax eigenvector for $\mathbf{e}_{L'}$ and the MDDBmax eigenvector for $\mathbf{e}_{N'}$. For comparison, the red curves in Figures 3f–3h show, respectively, the *L*, *M*, and *N* components of the magnetic field averaged over the four spacecraft and averaged over the time interval. For these curves, the coordinate directions are constant and the variation is because of the variation of the average magnetic field in the different intervals. The blue and red curves are fairly similar

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in Figures 3f–3h except at the smallest and largest time intervals. The values of B_L and B_{L^*} are roughly constant at about 12 nT because the center of our time intervals is roughly at the time when the magnetic field is its average between the magnetosphere and magnetosheath values (at the midpoint in time between the two green vertical lines in Figure 1b). The values of B_M and B_{M^*} have a larger variation but remain negative between −15 nT and −8 nT. These values of the guide field have larger magnitude than those from the later electron diffusion region event at 1307 UT (Denton et al., 2016b).

The variations in the normal field components, B_N (red) and B_{N^*}* (blue), are shown in Figure 3h. The two curves are in good agreement in the nest range $0.5 < \Delta t_{\text{range}}/\Delta t_0 < 2$, which includes a change of sign of the normal field component for the larger nest sizes. The positive values for the shorter nests are consistent with the direction of the plasma ejection jet (see Figure 1f) provided that the reconnection X point was located northward (at a larger L value) of the MMS spacecraft, which were at $Z \sim -0.7\ R_E$. Because B_L is positive in the magnetosphere and negative in the magnetosheath, B_N is expected to be positive in order for the magnetosphere magnetic field to connect to the magnetosheath magnetic field in this outflow region. The values of B_N and B_{N^*} are both positive at the intermediate time intervals. Large time intervals may include data for time periods for which the N direction has rotated or where more complicated structures are included, and the small time intervals may contain insufficient information to provide meaningful results. The issue of negative B_N is discussed further in section 6.

4.1.5. Summary of Results for Different Intervals

Both the maximum eigenvalue direction for MVAB in Figure 2 and the maximum gradient direction for MDDB in Figure 3 are roughly constant for time intervals up to about $\Delta t_{\text{range}}/\Delta t_0 = 2$. The maximum gradient direction is most orthogonal to the maximum variance direction, indicating consistency between the two methods, for time intervals greater than $\Delta t_{\text{range}}/\Delta t_0 = 0.5$ (blue curve in Figure 3e). Values of B_{N^*} using our hybrid method (blue curve in Figure 3h) are small and positive, as expected for this reconnection event, for $\Delta t_{\text{range}}/\Delta t_0$ ranging between about 0.3 and 1.25. Time intervals in the larger half of this range are preferred, since they better represent the main part of the current sheet crossing and because they lead to lower statistical error. Based on these results, our choice of the default time interval Δt_0 (the 4.5 s interval for which $\Delta t_{\text{range}}/\Delta t_0 = 1$) appears to be good. This interval satisfies the three criteria we mentioned in section 2: large eigenvalue ratios, consistency of directions over a range of time intervals, and, at least to a certain extent, consistency of method as indicated by the rough orthogonality between $\mathbf{e}_{L'}$ and $\mathbf{e}_{N'}$. In the next section, we will show how results from other methods generally also support our *L-M-N* coordinate system.

4.2. Other Methods

4.2.1. Minimum Variance of B: Results

Although $\lambda_{\text{MVAB},2}/\lambda_{\text{MVAB},3}$ for the smaller time interval 1305:43.5 to 1305:48 UT was only 6.3, that is still significantly better than the value 2.5 found for the larger time interval, 1305:40 to 1306:09 UT. Using the smaller time interval, we find that the MVABmin eigenvector is $\mathbf{e}_{N^*} = (0.889, -0.112, -0.443)$. This is 14° off from the normal direction of the hybrid method described in section 4.1, for which the normal direction is $\mathbf{e}_N = (0.9210, 0.1160, -0.3718)$. The GSE event location at $(8.32, 8.50, -0.70)R_E$ is well on the afternoon side of the magnetopause so that a positive value of the GSE Y component of $\mathbf{e}_N$ = would be expected. This expectation is met for the hybrid normal, albeit with an unexpectedly low value, but is not met for the MVAB normal.

Figure 2 shows the effects of using the MVABmin eigenvector for the N direction for different time intervals. Figure 2d shows that the minimum variance direction varies greatly with respect to the interval used. Based on the results of section 4.1.3 and other methods to be described below, our hybrid N direction is likely to be accurate to less than 10°. Note that the best agreement between the MVABmin eigenvector and the hybrid method N direction occurs when the ratio $\lambda_{\text{MVAB},2}/\lambda_{\text{MVAB},3}$ is largest at $\Delta t_{\text{range}}/\Delta t_0$ values between 0.6 and 1 (Figures 2a and 2d).

The red curve in Figure 2e shows the angle between the MVABmin eigenvector and our N direction, $\theta_{\text{MVABmin-}N}$. The large variation in $\theta_{\text{MVABmin-}N}$ in Figure 2e is a further indication that MVABmin is not a reliable predictor of the normal direction for this event.

The blue curves in Figures 2f–2h show, respectively, the L^*, M^*, and N^* components of the magnetic field averaged over the four spacecraft and over the time interval, similar to Figures 3f–3h but for L^*, M^*, N^* constructed using the MVABmax eigenvector for $\mathbf{e}_{L^*}$ and the MVABmin eigenvector for $\mathbf{e}_{N^*}$. In other words, the blue curves show the magnetic field components that would be calculated in each interval if only MVAB were

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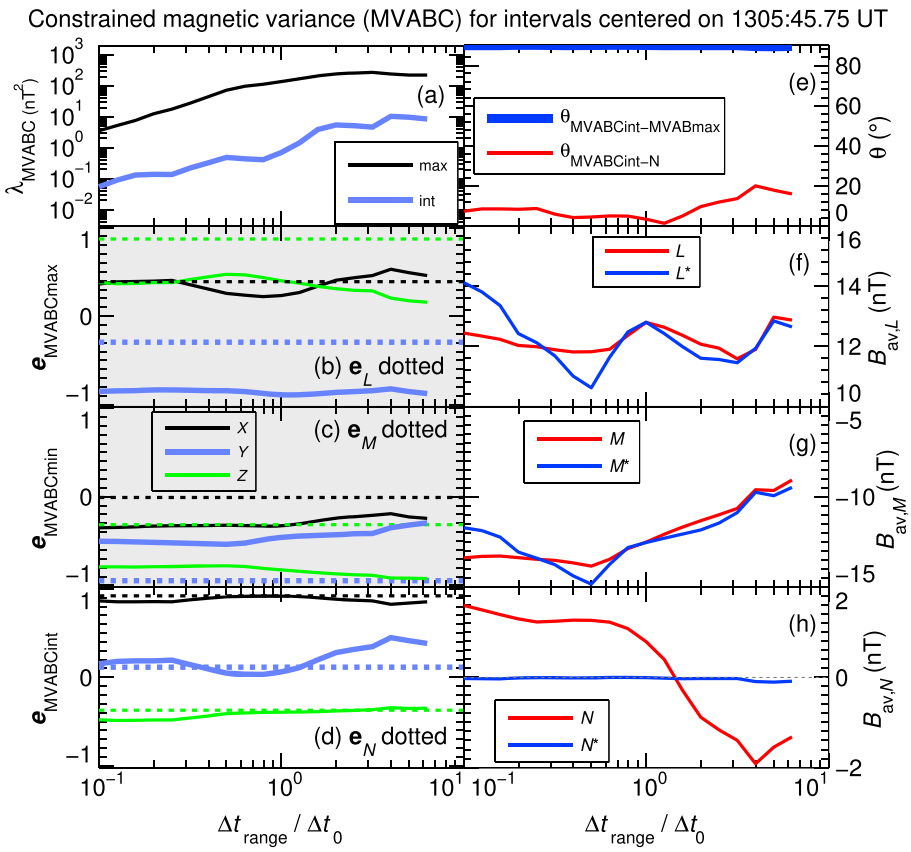

Figure 4. Results from constrained variance analysis of **B** (MVABC). Similar to Figure 2 but now showing (a) eigenvalues from MVABC. Direction cosines for the (b) maximum, (c) minimum, and (d) intermediate variance eigenvectors (curves) plotted along with the direction cosines of our *L*, *M*, and *N* directions, respectively (horizontal dotted lines). (e) Angles between the intermediate variance (MVABCint) eigenvector and the MVABmax eigenvector (blue curve) or our *N* direction (red curve). Time-averaged interval-dependent (f) *L*, (g) *M*, and (h) *N* components calculated using our fixed coordinate system (red curves) or an interval-dependent L^*-M^*-N^* coordinate system combining the MVABmax eigenvector for $\mathbf{e}_{L'}$ with the MVABCint eigenvector for $\mathbf{e}_{N'}$ (blue curves).

used to determine the coordinate directions. The red curves in Figures 2f–2h show, respectively, the *L*, *M*, and *N* components, just like in Figure 3 and later figures. Since we continue to use the MVABmax eigenvector to determine $\mathbf{e}_{L'}$, the blue and red curves in Figure 2f are fairly similar because the interval-dependent MVABmax eigenvector is consistently close to our *L* direction (see the dotted red curve in Figure 2e).

However, the B_{M*} and B_{N*} components, plotted, respectively, in Figures 2g and 2h (blue curves), are significantly different from the B_M and B_N components from the hybrid method (red curves). As discussed in section 4.1.4, the normal component of the magnetic field is expected to be positive at this time because the southward directed plasma jet indicates that the spacecraft are southward of the X point. But B_{N*} is negative (blue curve in Figure 2h). This again suggests that use of the MVABmin eigenvector for *N* is problematic for these intervals.

4.2.2. Constrained Minimum Variance of B: Results

As in many previous magnetopause studies, our analysis leads to a very small normal component of the magnetic field. This result suggests that the constrained magnetic variance method, MVABC (see section 3.1), could, on its own, provide an accurate determination of the normal direction for comparison with $\mathbf{e}_N$, the normal from our hybrid model. The price paid is that MVABC provides no information about the actual sign and magnitude of the normal magnetic field component.

Figure 4 shows results from MVABC. Since the direction of the average magnetic field, $\langle\mathbf{B}\rangle_t$, has been projected out of the $\mathbf{M}^{\mathrm{MVABC}}$ matrix, the minimum eigenvalue is automatically zero, and the minimum eigenvalue direction (Figure 4d) is automatically that of $\langle\mathbf{B}\rangle_t$. The intermediate eigenvalue (blue curve in Figure 4a) is well separated from the maximum eigenvalue (black curve), especially for small and intermediate time intervals.

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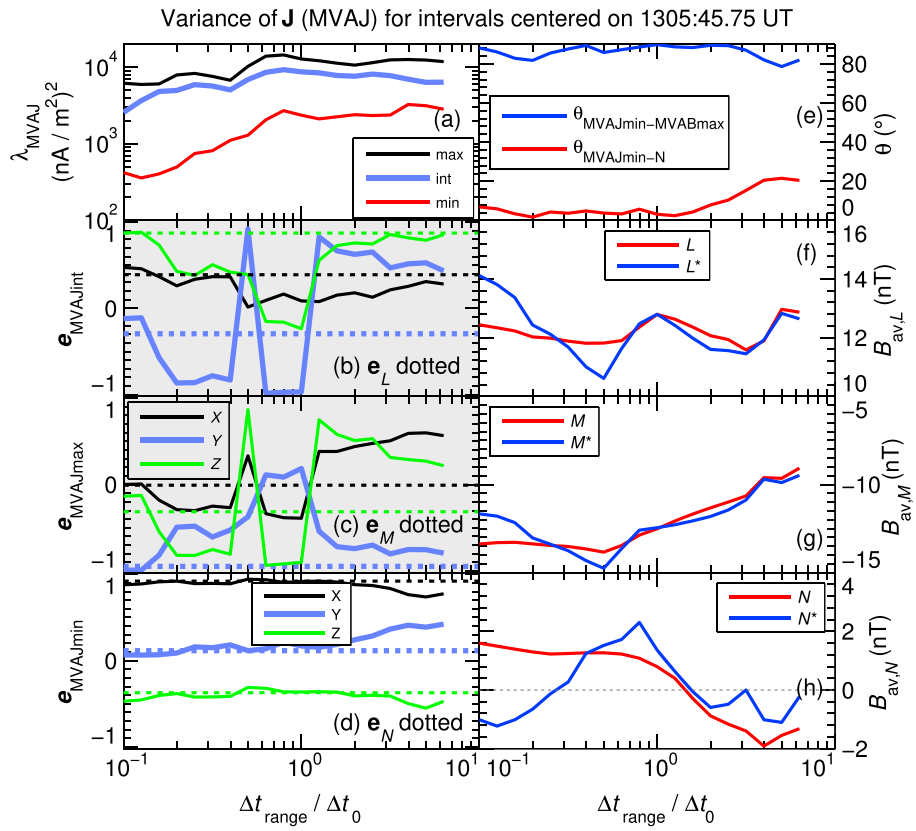

Figure 5. Results from variance analysis of **J** (MVAJ). Similar to Figure 2 but now showing (a) eigenvalues from MVAJ. Direction cosines for the (b) intermediate, (c) maximum, and (d) minimum variance directions (curves) plotted along with the direction cosines of our *L*, *M*, and *N* directions, respectively (horizontal dotted lines). (e) Angles between the minimum variance (MVAJmin) eigenvector and the MVAB maximum variance eigenvector (blue curve) or our *N* direction (red curve). Time-averaged interval-dependent (f) *L*, (g) *M*, and (h) *N* components calculated using our fixed coordinate system (red curves) or an interval-dependent L^*-M^*-N^* coordinate system combining the MVAB maximum variance eigenvector for $\mathbf{e}_{L'}$ with the MVAJmin eigenvector for $\mathbf{e}_{N'}$ (blue curves).

For those intervals, the intermediate eigenvalue direction agrees well with our $\mathbf{e}_N$ (Figure 4d and red curve in Figure 4e), whereas no agreement is seen in Figures 4b and 4c. The time and spacecraft-averaged **B** components (Figures 4f–4h) are similar to those in Figure 3, except that here B_{av,N^*} is very close to zero because the intermediate eigenvector is orthogonal to $\langle \mathbf{B} \rangle_t$ by construction. (B_{av,N^*} is not exactly zero because of the adjustment of $\mathbf{e}_{N'}$ to make $\mathbf{e}_{N^*}$ perpendicular to $\mathbf{e}_{L^*}$.)

4.2.3. Variance of J: Results

Figure 1d shows that $J_{\mathrm{av},N}$ found using $\mathbf{e}_N$ from the hybrid method (red curve) varies much less than the other components, especially in the short time interval between the green vertical lines. This behavior supports the idea of using MVAJmin, the minimum variance eigenvector from MVAJ, as a predictor of $\mathbf{e}_{N'}$, as discussed in section 3.1.

Figure 5 shows the results. The MVAJmin direction (Figure 5d) is roughly constant for time intervals up to about $\Delta t_{\mathrm{range}}/\Delta t_0 = 2$ and is vastly more constant than the other variance directions (Figures 5b and 5c). (There is a 180° ambiguity in the direction of the eigenvectors. The sign of the direction cosines of the maximum and intermediate variance eigenvectors plotted in Figures 5b and 5c is chosen so that the dot product between the maximum and intermediate variance eigenvectors and $\mathbf{e}_L$ and $\mathbf{e}_{M'}$ respectively, is positive. Because these variance directions are changing greatly, there are abrupt jumps in the curves.) Figures 5d and 5e show that the minimum variance direction is within 7.6° of the *N* direction of our hybrid method for intervals up to $\Delta t_{\mathrm{range}}/\Delta t_0 = 2$ (red curve in Figure 5e). The agreement is better for the smaller intervals for which $\lambda_{\mathrm{MVAJ},2}/\lambda_{\mathrm{MVAJ},3}$ is larger (Figure 5a); for $\lambda_{\mathrm{MVAJ},2}/\lambda_{\mathrm{MVAJ},3}$ greater than 5, the minimum variance direction is within 5.6° of the *N* direction of our hybrid method. The average magnetic field values (Figures 5f–5h)

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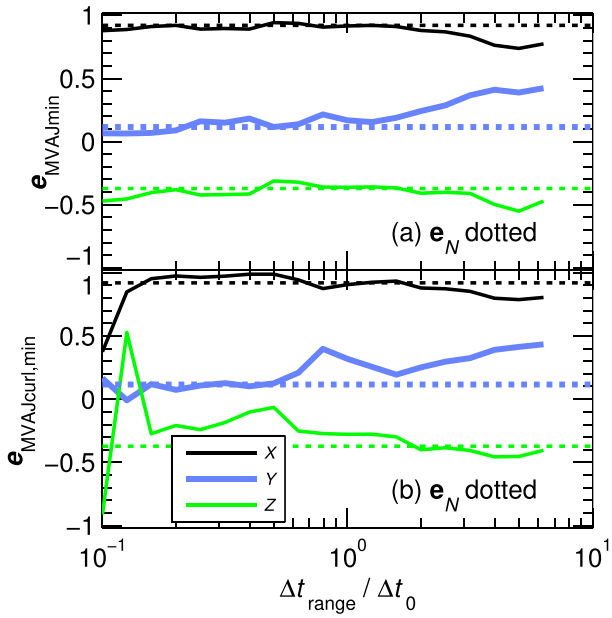

Figure 6. Minimum variance directions using the current density from the particle data and curlometer technique. Direction cosines to the *X*, *Y*, and *Z* Geo-Solar Ecliptic axes for the minimum eigenvalue directions of (a) $\mathbf{e}_{\mathrm{MVAJ,min}}$ calculated using **J** from the particle data and (b) $\mathbf{e}_{\mathrm{MVAJcurl,min}}$ calculated using $\mathbf{J}_{\mathrm{curl}}$ from the curlometer technique based on the magnetic field measurements. The direction cosines of the *N* direction using the hybrid method of this paper are indicated by dotted horizontal lines.

are very similar to those in Figure 3, again indicating good agreement with results from our hybrid method using the MDDBmax eigenvector for $\mathbf{e}_{N'}$.

We mentioned briefly in section 2 that use of **J** calculated from the particle moments led to more consistent results than use of $\mathbf{J}_{\mathrm{curl}}$ calculated from the magnetic field data. This point is illustrated in Figure 6. Figure 6a is the same as Figure 5d, showing the minimum variance direction of **J**. Figure 5b shows the minimum variance direction of $\mathbf{J}_{\mathrm{curl}}$, which clearly is not as constant as that of **J** (Figure 6a) with respect to the interval used. We do not know why this is. There are better statistics using **J**, since **J** can be calculated for each spacecraft, whereas $\mathbf{J}_{\mathrm{curl}}$ must be calculated from all the spacecraft data. Another possible reason relates to possible inaccuracy in the calculation of $\mathbf{J}_{\mathrm{curl}}$ due to nonuniformity of gradients.

4.2.4. Variance of J × B: Results

Figure 1e shows that $(J \times B)_{\mathrm{av},N}$, found using $\mathbf{e}_N$ from the hybrid method (red curve), varies more than the other components, especially in the short time interval between the green vertical lines. This behavior supports the idea that we can use the maximum variance direction of the magnetohydrodynamic force density, $\mathbf{J} \times \mathbf{B}$, as an estimator for $\mathbf{e}_{N'}$, as discussed in section 3.1.

Figure 7 is similar to Figure 2, except that the variance is that of $\mathbf{J} \times \mathbf{B}$ and that we use MVAJxBmax, the maximum variance direction from MVAJxB, as a predictor of $\mathbf{e}_{N'}$. This direction is nearly orthogonal to the MVABmax eigenvector (blue curve in Figure 7e). There is good agreement between the MVAJxBmax eigenvector and the *N* direction from our hybrid method (red curve in Figure 7e), especially at the smaller time intervals for which the separation in eigenvalues is greater (Figure 7a). Figure 7e shows that the MVAJxBmax eigenvector is within 6.5° of the *N* direction of our hybrid method for intervals up to $\Delta t_{\mathrm{range}}/\Delta t_0 = 1.65$. For $\lambda_{\mathrm{MVAJ,1}}/\lambda_{\mathrm{MVAJ,2}}$ greater than 6.8, the maximum variance direction is within 5.9° of the *N* direction of our hybrid method. And again, the average magnetic field values (Figures 7f–7h) are very similar to those in Figure 3.

4.2.5. Directional Derivative (MDD) Analysis: Results

We already discussed the MDDBmax eigenvector in section 4.1.3. We might well expect that the minimum gradient direction would be the out of plane direction $\mathbf{e}_M$, but Figure 3b shows that the minimum gradient direction is closest to $\mathbf{e}_L$, a result that was also noted by Denton et al. (2016a) for the 1307 UT time interval.

MDD can be used with any vector field observed by the four spacecraft, so we also tried using it with other quantities such as the ion velocity, electron velocity, electric field, and current density calculated from the particle data, **J**. The results were not as consistent or clear for these quantities. For all of them, as was the case using **B**, the minimum gradient direction was closer to $\mathbf{e}_L$ than to $\mathbf{e}_M$, indicating that the particle quantities were most constant along the direction of the dominant magnetic field.

4.2.6. MFR: Results

Figure 8 shows results for MFR using a format similar to that of Figure 2. The relevant direction for this method is the minimum eigenvalue (MFRmin) direction, which should correspond to the *N* direction, as discussed in section 3.3; so we use the MFRmin eigenvector for $\mathbf{e}_{N'}$. Although the MFRmin eigenvector is not constant with respect to the time interval used (Figure 8d), Figure 8e shows that there are some time intervals for which it is within 10° of our $\mathbf{e}_N$ (red curve in Figure 8e). In particular, this is true for the small time intervals for which the eigenvalue ratio $\lambda_{\mathrm{MFR,2}}/\lambda_{\mathrm{MFR,3}}$ is the largest (Figure 8a). The L^* and M^* components of the average magnetic field (blue curves in Figures 8f and 8g) are very similar to the *L* and *M* components using our hybrid method (red curves), but the N^* component is significantly different from that of *N* (Figure 8h). There is a range of positive values for B_{N^*} for the intermediate time intervals, but the magnitude of B_{N^*} is significantly larger than that of B_N.

4.2.7. Summary of Results From the Other Methods

Though none of these other methods has as great a separation of eigenvalues or as consistent a direction as the MVABmax and MDDBmax eigenvectors used in our hybrid method, all of them yield a normal direction that is close to that of our hybrid method if intervals with large eigenvalue ratios are used. For this set of intervals, the best of these other methods, judged by consistency of results, seems to be variance analysis

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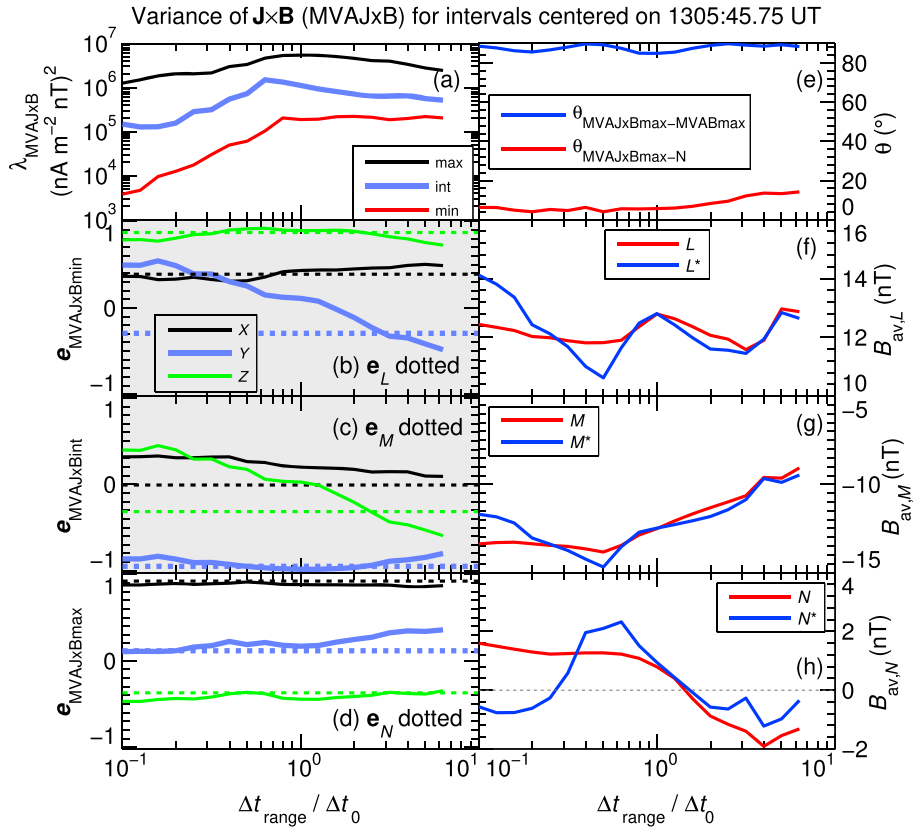

Figure 7. Results from variance analysis of $\mathbf{J} \times \mathbf{B}$ (MVAJxB). Similar to Figure 2 but now showing (a) eigenvalues from MVAJxB. Direction cosines for the (b) minimum, (c) intermediate, and (d) maximum variance directions (curves) plotted along with the direction cosines of our *L*, *M*, and *N* directions, respectively (horizontal dotted lines). (e) Angles between the maximum variance (MVAJxBmax) eigenvector and the MVABmax eigenvector (blue curve) or our *N* direction (red curve) and time-averaged interval-dependent (f) *L*, (g) *M*, and (h) *N* components calculated using our fixed coordinate system (red curves) or an interval-dependent L^*-M^*-N^* coordinate system combining the MVABmax eigenvector for $\mathbf{e}_{L'}$ with the MVAJxBmax eigenvector for $\mathbf{e}_{N'}$ (blue curves).

using $\mathbf{J}$ or $\mathbf{J} \times \mathbf{B}$; the results shown in Figures 5 and 7 for these methods were very similar to those shown in Figure 3 using the MDDBmax eigenvector for $\mathbf{e}_{N'}$.

Table 3 examines numerically the consistency of the different directions that we have used to get $\mathbf{e}_{N'}$ for time intervals $\Delta t_{\text{range}}/\Delta t_0 = 0.5$ to 2 (seven intervals). While consistency with respect to the time interval is not necessarily a direct indicator of accuracy, lack of consistency is definitely an indicator of uncertainty. The left column lists the various methods that we have examined. The middle column shows the mean value, averaging over the different intervals, of the angle of deviation between the interval-dependent $\mathbf{e}_{N'}$ and the direction of $\mathbf{e}_{N'}$ averaged over the different time intervals. Ultimately, the more important direction may be $\mathbf{e}_{N*}$, found through combination of $\mathbf{e}_{N'}$ from the individual method and $\mathbf{e}_{L'}$ from the MVABmax eigenvector. The last column in Table 3 shows the mean angle between the interval-dependent $\mathbf{e}_{N*}$ and the direction of $\mathbf{e}_{N*}$ averaged over the different time intervals. These measures of consistency are best (smallest deviation from the mean) for MDDBmax, MVAJmin, and MVAJxBmax.

5. Magnetopause Models: Results

Here we compare the *N* direction from our hybrid method to the normal directions from the magnetopause models of Shue et al. (1998) and Lin et al. (2010). At 1305:45, the position of the MMS spacecraft was $(X, Y, Z) = (8.32, 8.50, -0.70)$. Using 1 min data from NASA/Goddard Space Flight Center's OMNI data set through

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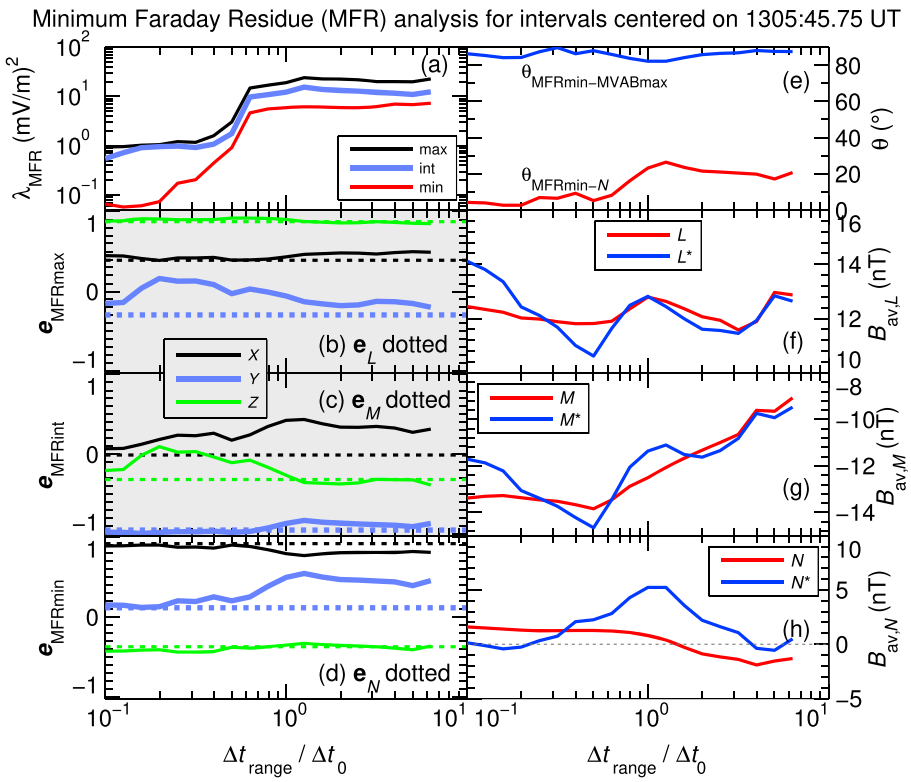

Figure 8. Results from minimum Faraday residue analysis (MFR). Similar to Figure 2 but now showing (a) MFR eigenvalues, MFR eigenvectors (b–d), (e) angles between the MFR minimum eigenvalue (MFRmin) eigenvector and the MVABmax eigenvector (blue curve) or our *N* direction (red curve). Time-averaged interval-dependent (f) *L*, (g) *M*, and (h) *N* components calculated using our fixed coordinate system (red curves) or an interval-dependent L^*-M^*-N^* coordinate system combining the MVABmax eigenvector for $\mathbf{e}_{L'}$ with the MFRmin eigenvector for $\mathbf{e}_{N'}$ (blue curves).

OMNIWeb (King & Papitashvili, 2005), the solar wind proton density was 7.80 cm^{-3}; the solar wind dynamic pressure was 1.35 nPa; the interplanetary magnetic field, (B_X, B_Y, B_Z), was (3.54, 2.56, 1.55) nT; and the solar wind velocity, (V_X, V_Y, V_Z), was (−456.8, 15.8, −5.9) km/s.

The Shue et al. (1998) normal direction projected back (along the normal) to the position of the spacecraft was (0.860, 0.508, −0.042). This is 30° off from the normal direction from the hybrid method. The Shue et al. (1998) model is symmetric around the Earth-Sun axis (*X* axis). But the normal from the hybrid method has an angular position about the *X* axis that is far from that of the MMS spacecraft.

Table 3
Mean Angle of Deviation From the Average Direction[a]

Direction used for $\mathbf{e}_{N'}$	Mean deviation using $\mathbf{e}_{N'}$ (deg)	Mean deviation using $\mathbf{e}_{N^*}$ [b](deg)
MVABmin	15.4	15.4
MVABCint	4.3	4.3
MVAJmin	2.6	2.0
MVAJxBmax	2.2	2.1
MDDBmax	2.1	2.0
MFRmin	6.8	6.7

Note. MVABmin = MVAB minimum variance direction; MVABCint = constrained minimum variance direction perpendicular to the average magnetic field; MVAJmin = minimum variance direction for J; MVAJxBmax = maximum variance direction for J x B; MDDBmax = maximum directional derivative direction; MFRmin = MFR minimum eigenvalue direction.
[a]Average over intervals $\Delta t_{range}/\Delta t_0$ = 0.5 to 2. [b]$\mathbf{e}_{N^*}$ is calculated for each interval combining the direction used to get N' with the MVABmax direction for L' using the method of Appendix A.

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This result led us to consider the Lin et al. (2010) model, which is not symmetric around the Earth-Sun line and includes the effective dipole tilt. In this case, we calculated the magnetopause normal direction using the angular position of the spacecraft position vector. The Lin et al. (2010) normal direction was (0.839, 0.495, 0.228). This is 42° off from the normal direction from the hybrid method. So using the asymmetric model did not improve the agreement. The conclusion is that the magnetopause models do not very well represent the observed current sheet orientation. This is evidence for a large local deformation of the magnetopause, such as could result from a "lower-hybrid drift-like instability" (Ergun et al., 2017; Le et al., 2017; Price et al., 2017), the Kelvin-Helmholtz instability or a plasma/field structure from the solar wind impinging on the magnetopause.

6. Summary

Our goal in this paper has been to determine a coordinate system, *L-M-N*, for reconnection events observed by the MMS spacecraft at the magnetopause or elsewhere. At the magnetopause, the *L* axis (northward) is defined to be along the reconnecting component of the magnetic field, the *N* axis (sunward) is along the direction normal to the reconnecting current sheet, and the *M* axis (westward) completes the right-handed orthogonal system.

In the past, when the data available mostly came from a single spacecraft, the most common approach was to use variance analysis of the magnetic field (MVAB), with *L*, *M*, and *N* corresponding to the directions of maximum, intermediate, and minimum variance, respectively. To be successful, this method requires a data interval that contains a substantial portion of the current sheet traversal. It often provides a reliable *L* direction because the maximum and intermediate eigenvalues of the variance matrix are often extremely well separated. But it can give questionable results for *M* and *N* when the two smallest eigenvalues of the variance matrix are not well separated, which is often the case.

The recent MMS mission, with its four, very closely spaced observation platforms, has opened up an array of new possibilities to circumvent this problem, especially regarding the determination of *N*, which can now be predicted (sometimes after small adjustments) as the direction of maximum gradient of the magnetic field, provided that the ratio of maximum to intermediate eigenvalue is sufficiently large. And the gradient can, if necessary, be calculated at the cadence of the magnetic field measurements. This method is referred to as MDDB (section 3.2 and Shi et al., 2005) but the gradient of other quantities can also be used. The "hybrid" method we have focussed on here is a combination of MDDB and MVAB. It takes the *M* direction to be perpendicular to the plane containing the direction of the maximum gradient of **B** (MDDBmax), $\mathbf{e}_{N'}$, and the direction of maximum variance of **B** (MVABmax), $\mathbf{e}_{L'}$, and then makes usually small rotations of $\mathbf{e}_{L'}$ and $\mathbf{e}_{N'}$ around the *M* axis to form an orthogonal pair, L^* and N^* (see Appendix A). For the hybrid method using MVABmax and MDDBmax, these adjusted directions become our *L* and *N*.

The *L-M-N* system obtained from this hybrid procedure has been compared to those from a variety of other methods (section 3). Among these is the set of three eigenvectors from MVAB on its own. Included also is a version (MVABC) constrained so that the average normal magnetic field component is zero. Also presented are eigenvectors from variance analysis of the measured electric current density **J** (MVAJ) and of the electromagnetic force $\mathbf{J} \times \mathbf{B}$ (MVAJxB) and, finally, MFR analysis. After this study was completed, we became aware of another recent study that introduced a method called Local Normal Analysis (LNA) (Rezeau et al., 2018). This new method may also be useful; like MDD, it can provide a point by point evaluation of the normal direction.

The principal event for the study is a magnetopause crossing on 16 October 2015, during the 4.5 s time interval 1305:43.5–1305:48.0 UT, preceding the encounter with the electron diffusion region around 1307:02 UT reported by Burch et al. (2016). The chosen interval proved optimal for our hybrid method to obtain the *L-M-N* system but not necessarily for all of the other methods, with which we compare our prediction for the all important normal direction *N*.

The results of these comparisons are shown for the 4.5 s interval in Figure 9a. This polar plot, oriented with its polar axis (origin in the plot) along the positive GSE *X* axis, also shows a blue curve separating regions of positive and negative average normal magnetic field components and a green curve, along which the normal vectors are exactly perpendicular to the uncorrected maximum variance direction (L') from MVAB. The normal vector from our hybrid method is shown as a black dot. The unadjusted normal directions (N') from several methods using the 4.5 s interval are shown by colored letters, while the black letter "B" shows the *N* direction from the MVAB direction used by Burch et al. (2016). The figure also shows the directions of the

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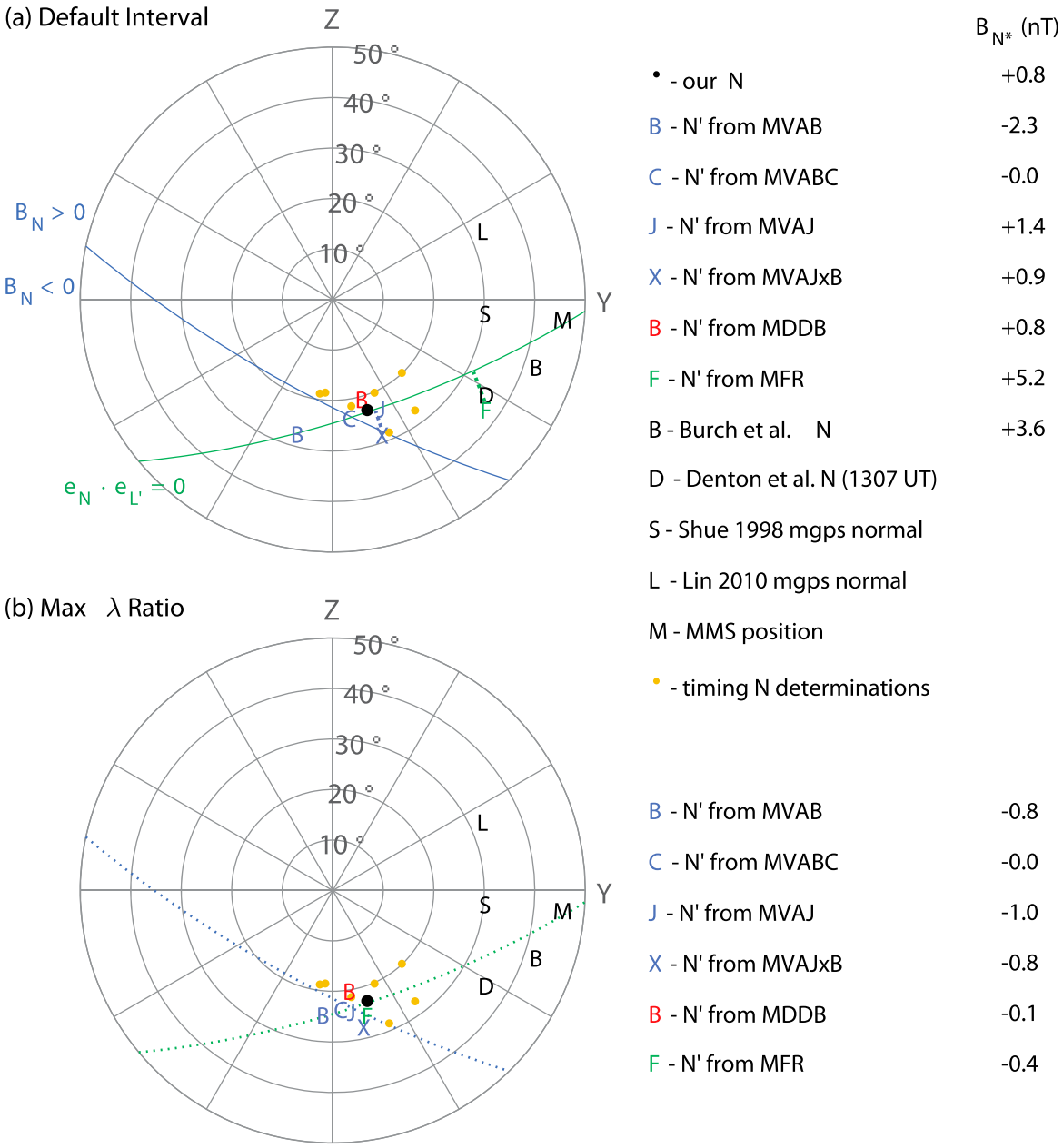

Figure 9. Normal directions and field components. (a) Polar plot with normal directions for the 4.5 s interval, with polar angle from the Geo-Solar Ecliptic (GSE) *Y* direction (rightward) toward the GSE *Z* direction (upward) and polar radius representing the angle from the GSE *X* direction. The black dot shows our *N* direction from the hybrid method; the blue letters show the unadjusted *N′* directions using variance analysis; the red "B" shows the *N′* direction from the direction derivative of **B**; the green "F" shows the *N′* direction from minimum Faraday residue analysis (MFR); the black "B" shows the *N* direction of Burch et al. (2016) using a larger time interval; the black "S" and "L" show the normal directions for the magnetopause (mgps) models, and the black "M" shows the angular position of MMS; the gold dots are from timing analysis. The position of each letter symbol in the plot is at the center of the smallest rectangle that would surround that letter. The blue curve marks the arc perpendicular to the average **B**, and the green curve marks the arc perpendicular to our unadjusted *L′* direction from MVABmax. After adjusting the *N′* directions to be perpendicular to the adjusted *L′* direction, the resulting *N** directions move closer to the green curve as indicated by the dotted blue and green lines. (b) Same as (a) except that the *N′* determinations (blue, red, and green letters) are found for each method using the nested time interval with the greatest eigenvalue ratio.

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Shue et al. (1998) and Lin et al. (2010) magnetopause normals as the letters "S" and "L," respectively, and a letter "M" indicating the orientation of the MMS position vector. The position of each letter symbol in the plot is at the center of the smallest rectangle that would surround that letter if such a rectangle were drawn. The exact directions of the N' and N^* estimates are listed in supporting information Table S2. The B_{N^*} value for each method is shown in the legend.

All of our determinations of the normal direction in Figure 9a are located in the low part of the diagram, far away from the model normals. We conclude that there must have been a large, spatially and temporally localized, deformation of the magnetopause shape during the event, causing a large deviation of the predicted normal direction from the Shue and Lin models. The physical reason for this behavior remains unclear but is possibly related to localized reconnection activity in combination with a lower-hybrid drift-like instability (Ergun et al., 2017; Le et al., 2017; Price et al., 2017), large-amplitude Kelvin-Helmholtz waves, or a plasma/field structure from the solar wind impinging on the magnetopause.

As indicated in the legend on the right, the normal direction from the hybrid method, as well as those from MVAJ, MDDB, and MFR, all indicate positive normal magnetic field components, which is in agreement with the observed southward direction of the reconnection jet, indicating a spacecraft location south of the reconnection site. On the other hand, the normal from MVAB has a substantial negative component (presumably a consequence of insufficient separation of the two smallest eigenvalues), which is inconsistent with the jet direction and therefore is judged to be spurious.

The $B_{N'}$ component from MVAJxB, while positive, is very small. This behavior could be the result of an expected bias when the method is applied to reconnection events, since they have a nonzero normal magnetic field component so that the $\mathbf{J} \times \mathbf{B}$ force has a nonnegligible and nonconstant component parallel to the current sheet. This bias causes the predicted normal vector in Figure 9a to lie nearly on the blue curve. The normal also deviates considerably from being on the green line. Adjustment of the N' direction to be perpendicular to the adjusted L' direction, by rotating $\mathbf{e}_{N'}$ about $\mathbf{e}_M$ using the method of Appendix A, moves the resulting N^* direction along the dotted blue line close to the green curve, where B_{N^*} is positive.

The MVABC normal lies exactly on the blue curve, as required, but also happens to be nearly on the green curve so that it is nearly perpendicular to the maximum variance direction of the magnetic field.

The reason that the MFR normal is so far from the other normals, and also so far from the green curve, appears to be that the 4.5 s interval is not optimal for this method. The MFR results change significantly depending on the interval chosen (see Figure 8d). Adjusting $\mathbf{e}_{N'}$ so that $\mathbf{e}_{N^*}$ is perpendicular to $\mathbf{e}_{L^*}$ moves the green "F" in Figure 9a along the green dotted line to a position close to the green solid curve. The new direction of $\mathbf{e}_{N^*}$ gives an average normal field component during the 4.5 s interval that remains positive but is probably still too large.

Since the 4.5 s time interval was not optimal for all of the methods, we also plot in Figure 9b the directions $\mathbf{e}_{N'}$ (blue, red, and green letters) determined for the different methods, where for each method we use the time interval for which the relevant eigenvalue ratio is the largest. In this case, the determinations of $\mathbf{e}_{N'}$, including that of MFR, are clustered closer together, slightly lower and to the left of the black dot (the normal from our hybrid method using the 4.5 s interval). Note that the green and blue curves, now dashed, refer only to the black dot. The other methods have individual such curves (not shown), corresponding to their individual optimal data intervals. Note also that all methods, except for our hybrid normal, now have negative, albeit very small negative normal field components. This result is an indication that interval optimization on the basis of eigenvalue ratios does not always lead to more accurate results, especially when those intervals become very small. Nevertheless, the fact that the narrow cluster of new symbols in Figure 9b is close to the black dot (our optimal hybrid normal) lends support to our conclusion that the hybrid method can give very reliable results. Since MDDB uses the gradient of $\mathbf{B}$, this proximity is also an indication that the orientation of the MMS tetrahedron must be very accurate.

On the whole it appears that the hybrid normal (black dot in Figure 9) must be close to the true answer, at least for the portion of the crossing within the 4.5 s interval. We have estimated the statistical uncertainty as about 5°. The accuracy of our determination is supported by the close proximity of the black dot in Figure 9 to a cluster of normals from several of the other methods. As further evidence, we plot in Figure 9 the normal directions determined from timing analysis (Schwartz, 1998) at seven times during the 4.5 s interval. Timing analysis uses the times of events observed by all four spacecraft; in this case, we use the times of crossing

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of particular B_L values, as described in supporting information Text S2, Figures S1 and S2, and Table S2. We attempted to choose time intervals for which the slope of all four spacecraft curves was nonzero with roughly the same value over a range of time exceeding the time interval. Timing analysis assumes that the structure is static in the moving frame and totally one-dimensional, assumptions that are certainly not exactly correct. Nevertheless, the seven normals from the timing analysis (gold dots in Figure 9) are all within 10° of the hybrid method result (black dot in Figure 9), and the average direction from the seven timing analysis normals is only 1.8° from the hybrid method normal. Based on all these results, we estimate that the uncertainty of the normal direction within the 4.5 s interval is at least several degrees but probably less than 10°.

Nevertheless, it is also clear that persistent, and probably unavoidable, difficulties remain in obtaining accurate predictions of the size of the normal magnetic field component. As Figure 9b illustrates, even getting its correct sign can be difficult, unless the magnitude is sufficiently large. It appears clear that the normal field component for our event, while presumably positive, must have been small or even very small. An unexplained feature is the fact that our coordinate system leads to negative B_N in some of the magnetosheath interval (most of the interval between the rightmost green vertical line and $t = 61$ s in Figure 1b). The more consistently positive sign of B_N in Figure 1a was one reason that coordinate system was chosen for the Burch et al. (2016) study.

But there is evidence of time-dependent changes in the normal direction. Figure 1c indicates that the maximum eigenvector direction (MDDmax) was highly variable in the interval 47–64 s, and timing analysis results in this interval are much more scattered than for the 4.5 s interval (supporting information Text S2, Figures S1 and S2, and Table S2). The reasons for this behavior are not understood. Maybe conditions were sufficiently turbulent, on the scale of the spacecraft separation, so as to render meaningless the interpretation of MDDmax as a magnetopause normal. But there is also evidence in the magnetic field (Figures 1a or 1b) of some kind of larger-scale wave activity during this time interval. It is easily understandable that the more steady behavior between the green lines led to a much better MDDB eigenvalue ratio for this time interval than for the entire one shown in the figure.

In the supporting information Text S4 and Tables S4 and S5, we also examine a few other MMS events. Overall, our study indicates that our hybrid method often, but not always, produces superior results for the *L-M-N* coordinate system.

The *L*, *M*, and *N* directions we found for the 4.5 s interval near 1305:45 UT, obtained from the hybrid method, differ by 19°, 31°, and 24°, respectively, from those given by Denton et al. (2016a) for the current sheet near 1307:02 UT (compare the positions of the black dot and black "D" in Figure 9a). These large changes are present even though the time separation between the two crossings is only 77 s. In supporting information Text S3, Figures S3 and S2, and Table S3, we describe results from our hybrid method for five different intervals that suggest that the normal direction shifted in time first in the positive GSE *Y* direction toward and beyond the Burch et al. (2016) normal and then returned to the Denton et al. (2016a) normal at 1307:02 UT (Figure S2). While the coordinate system derived by Burch et al. (2016) was adequate for studying the normal at that time (compare the black D and B in Figure 9a), reexamination of the later 1307:02 UT interval shows that the coordinate system determined locally using the hybrid method (Denton et al., 2016a) better organizes the observed electric field, with E_N quasistatic, E_M, localized to the region of field line opening, and E_L accurately representing $E_{||}$, supporting the use of the hybrid method for that interval.

When considering time intervals for use in determining a coordinate system, the interval containing the region of the maximum gradient of **B** is a good one to start with. Looking for time intervals for which the eigenvalue separation is good, for which sensitivity to the choice of precise interval duration is small, for which different methods give a consistent result, and for which the normal component of the magnetic field has the expected sign and reasonable magnitude is an important means for obtaining an optimal coordinate system and for assessing its quality.

Appendix A: Defining *L-M-N* From Magnetic Variance and MDDB When the Eigenvalues Are Comparable

Suppose that the maximum variance eigenvector is $\mathbf{e}_{L'}$ and the MDDB maximum gradient eigenvector is $\mathbf{e}_{N'}$. Ideally, we want to use $\mathbf{e}_{L'}$ for $\mathbf{e}_L$ and $\mathbf{e}_{N'}$ for $\mathbf{e}_N$, but in general the two directions will not be exactly orthogonal.

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We want to create an *L-M-N* coordinate system with $\mathbf{e}_L$ and $\mathbf{e}_N$ exactly orthogonal but using a compromise between making $\mathbf{e}_L$ exactly equal to $\mathbf{e}_{L'}$ and making $\mathbf{e}_N$ exactly equal to $\mathbf{e}_{N'}$.

We can still define $\mathbf{e}_M$ as $\mathbf{e}_{N'} \times \mathbf{E}_{L'}$. Now we find the angle $\theta_{\mathrm{MVAB,MDDB}}$ from $\mathbf{e}_{N'}$ to $\mathbf{e}_{L'}$ and define $d\theta_{\mathrm{MVAB,MDDB}} \equiv \theta_{\mathrm{MVAB,MDDB}} - 90°$. Define $\lambda_{\mathrm{MVAB},1/2} \equiv \lambda_{\mathrm{MVAB},1}/\lambda_{\mathrm{MVAB},2}$, $\lambda_{\mathrm{MDDB},1/2} \equiv \lambda_{\mathrm{MDDB},1}/\lambda_{\mathrm{MDDB},2}$, and the direction perpendicular to $\mathbf{e}_{N'}$ in the plane perpendicular to $\mathbf{e}_M$, $\mathbf{e}_{L''} \equiv \mathbf{E}_M \times \mathbf{E}_{N'}$. To get $\mathbf{e}_N$, rotate $\mathbf{e}_{N'}$ toward $\mathbf{e}_{L''}$ an amount $d\theta_{N',N} = d\theta_{\mathrm{MVAB,MDDB}}\lambda_{\mathrm{MVAB},1/2}/(\lambda_{\mathrm{MVAB},1/2} + \lambda_{\mathrm{MDDB},1/2})$. That is, $\mathbf{e}_N = \cos(d\theta_{N',N})\mathbf{E}_{N'} + \sin(d\theta_{N',N})\mathbf{E}_{L''}$. Then complete the coordinate system with $\mathbf{e}_L$ as $\mathbf{e}_M \times \mathbf{E}_N$.

Note that if $\lambda_{\mathrm{MVAB},1/2} \ll \lambda_{\mathrm{MDDB},1/2}$, the rotation of $\mathbf{e}_{N'}$ toward $\mathbf{e}_{L''}$ will be small, yielding $\mathbf{e}_N \approx \mathbf{E}_{N'}$, but if $\lambda_{\mathrm{MVAB},1/2} \gg \lambda_{\mathrm{MDDB},1/2}$, the rotation will be almost the total angle $d\theta_{\mathrm{MVAB,MDDB}}$ needed to bring it 90° off from $\mathbf{e}_{L'}$, yielding $\mathbf{e}_L \approx \mathbf{E}_{L'}$.

Acknowledgments
Work at Dartmouth was supported by the NASA grant NNX14AC38G. H. H. was supported by the JSPS Grant-in-Aid for Scientific Research KAKENHI 15K05306. We thank R. L. Lin for help in implementing the Lin et al. (2010) magnetopause model. MMS data are available at https://lasp.colorado.edu/mms/sdc/public/links. More detailed results can be found in the supporting information; other data plotted in the figures of this paper are available from the first author by request.

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